

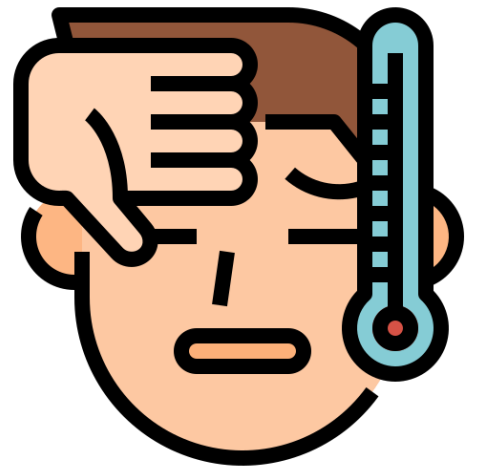


신약개발 단계별 AI 기술의 활용

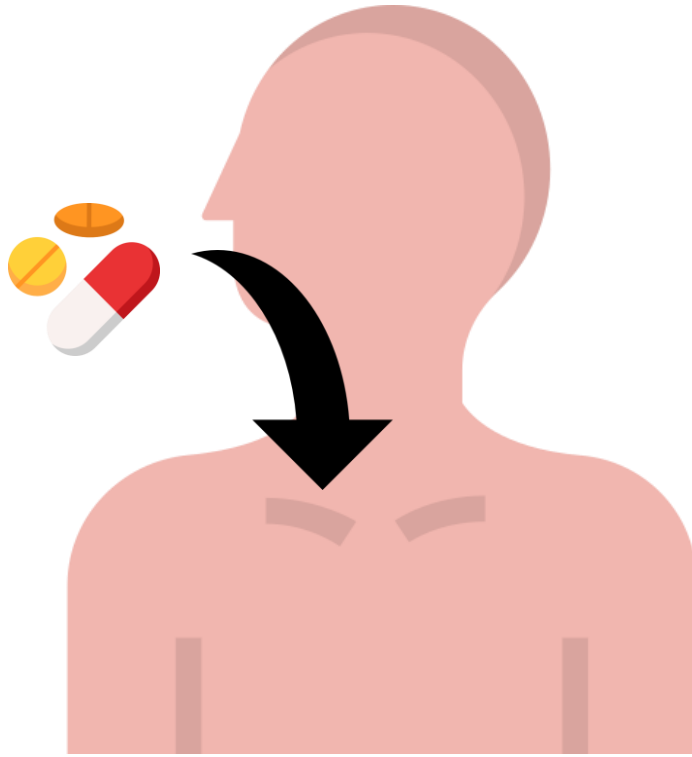
신현길

신약 개발에 AI가 왜 필요할까?

- 경제적인 문제 (개발에 지나치게 많은 비용과 시간이 든다)
 - 코로나 치료제 사례 (약이 상대적으로 빠르게 개발되었지만 치료제 사용까지 꽤 오랜 시간 소요)
- 희귀 질환 (소수의 환자군)
 - 경제적인 이유로 외면 받고 있는 환자들
- 동물 시험과 윤리
 - 안전성과 유효성 검증을 위해 희생되는 동물들



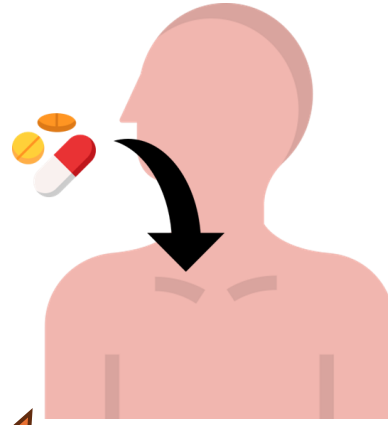
신약 개발은 왜 어려울까?



약을 혈액으로 전달하는 것이 목표
(약효를 낼 수 있는 충분한 양의 약물이 혈액에 있는지?)

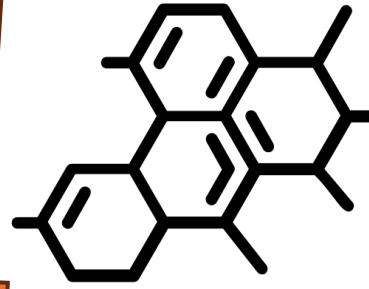
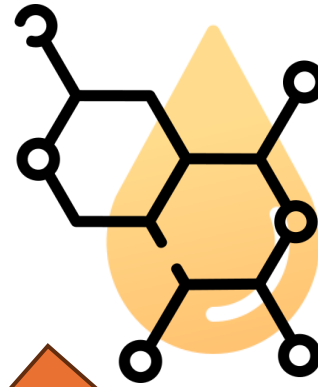
약물 흡수

물에 녹아야 약물 흡수
기름(세포막)에도 녹아야
세포 내로 유입



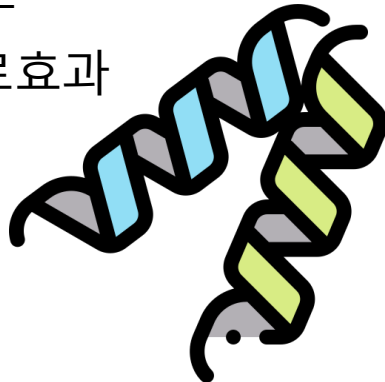
약물 대사

외부에서 유입된 물질을
제거하기 위한 간의 해독 작용



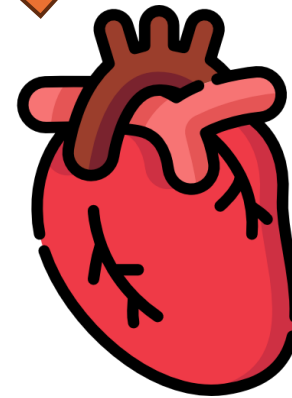
표적 단백질과 결합

질병 부위 세포 안에 있는
표적 단백질에 결합해서 치료효과



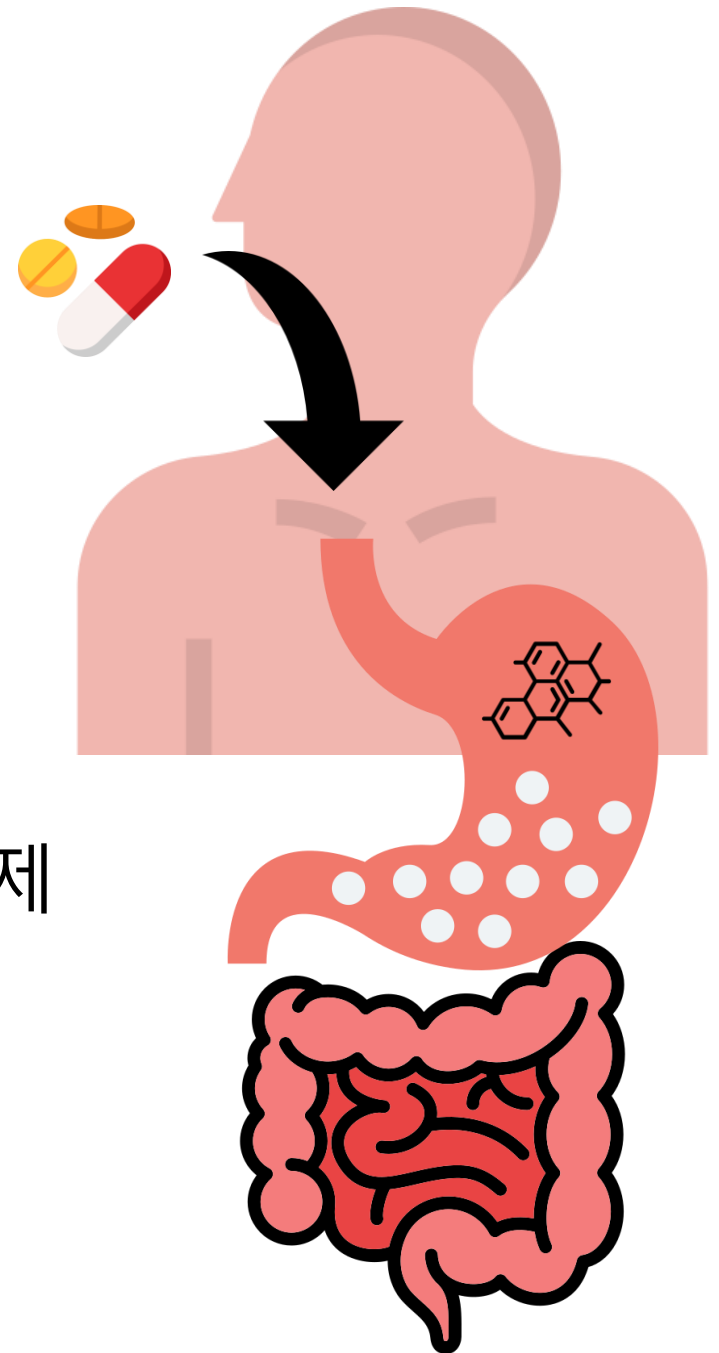
의도하지 않은 부작용

병변과 상관없는 장기에서
불필요한 작용을 일으키면
독성 유발



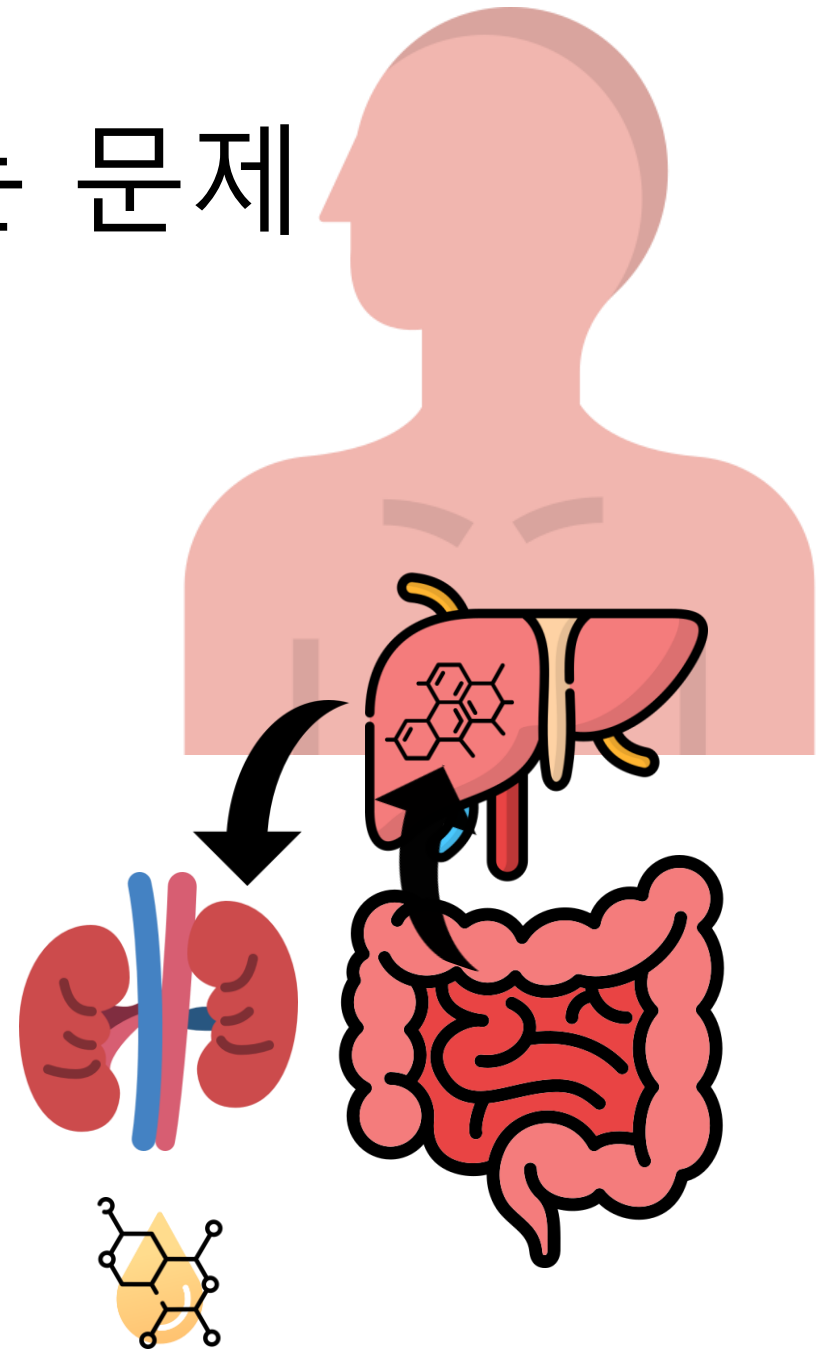
약이 물에 녹지 않는 문제

- 입으로 먹는 약은 합성 의약품
(화학 반응을 이용해 설계된 저분자 화합물,
small molecule)
- 합성 의약품은 용해도가 매우 낮은 것이 문제
 - 약이 물에 잘 녹아야 흡수 가능
 - 약이 물에 녹아야 다른 실험도 수행 가능



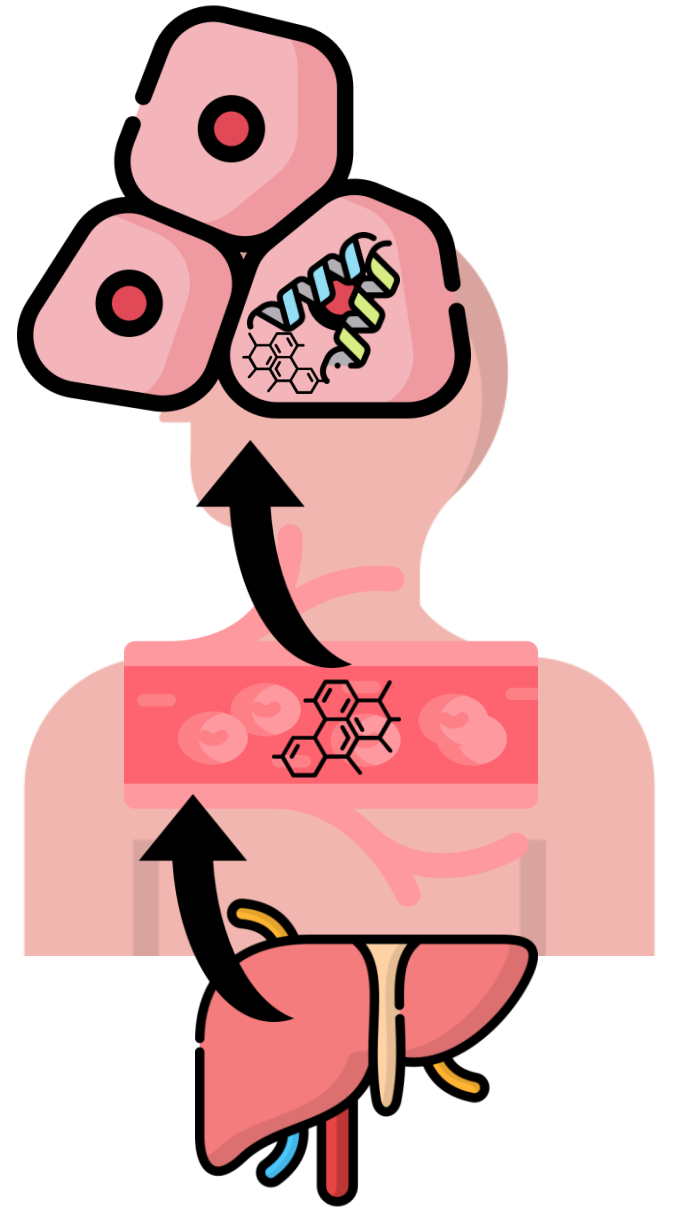
간을 거쳐 약물이 제거되는 문제

- 흡수된 약물은 간으로 이동
- 간에서 해독 작용 (약물 대사) 발생
 - 약물 대사는 외부에서 유입된 물질을 물에 더 잘 녹는 구조로 변화시키는 화학 반응 (소변을 통해 체외로 빠르게 배출)
 - 간에 발현된 다양한 효소에 의해 대사.



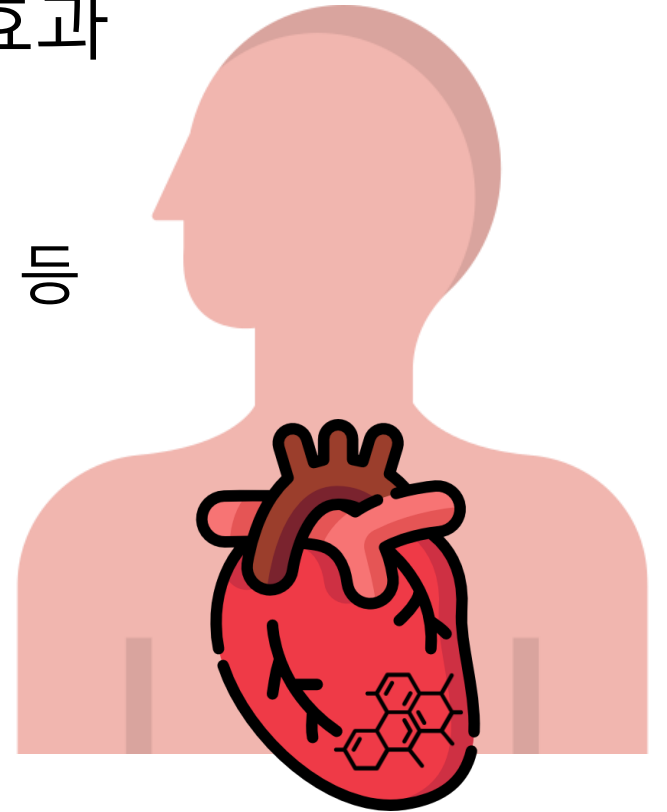
약물의 지용성 문제

- 질병 부위에 있는 세포로 약물이 유입
 - 세포막을 통과 (세포막은 기름)
 - 물에도 잘 녹고, 기름에도 잘 녹아야 약효가 발생
 - 세포 안에 있는 표적 단백질과 결합
- 약효를 나타내기 위해 **충분한 양의 약물**이 세포내로 전달되어야 함

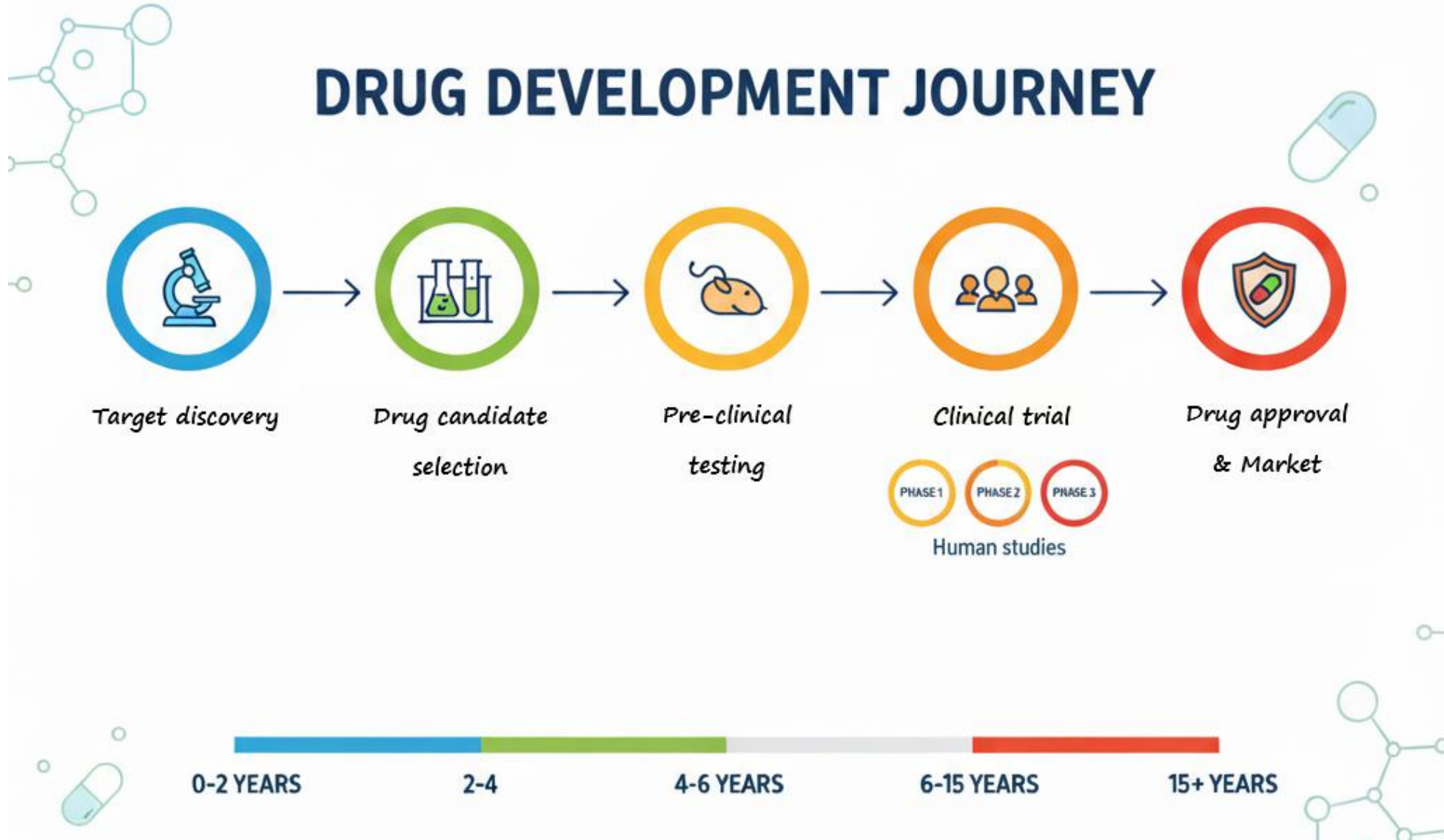


약물의 안전성 문제

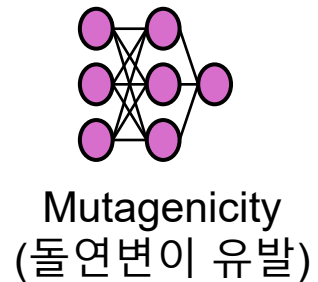
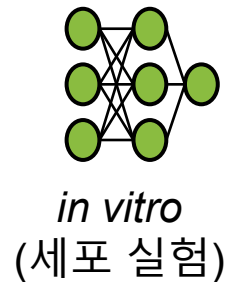
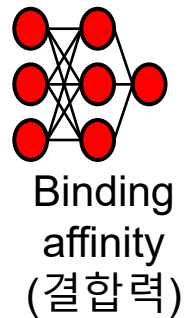
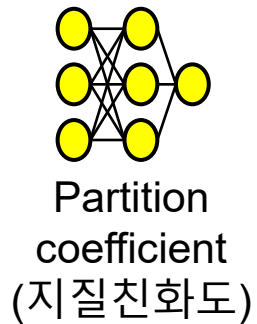
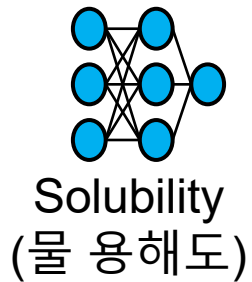
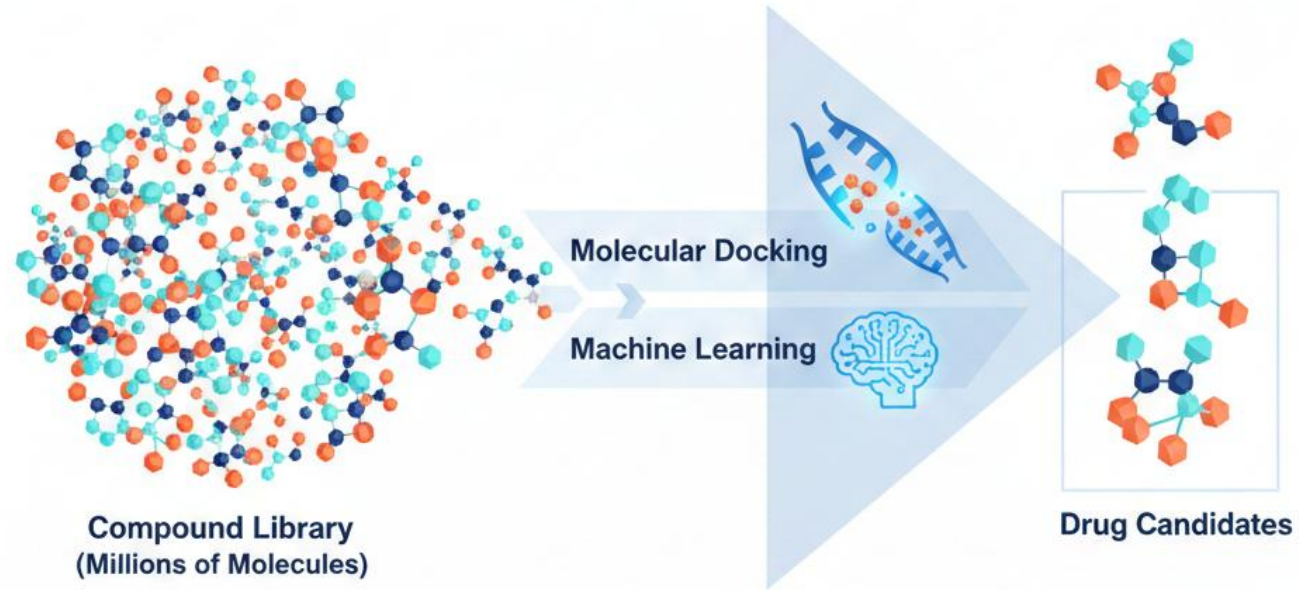
- 혈액을 통해 온 몸을 순환하면서 의도치 않은 효과
 - 약물에 의한 부작용 발생
 - 예: 심장 박동에 영향을 주는 부작용, 돌연변이 유발 등
- 부작용을 피하기 위한 방법
 - 부작용이 발생하는 기전을 이해하고 실험으로 확인
 - 낮은 농도에서 약효를 나타내는 물질 개발
(타겟 단백질에 선택적으로 강하게 결합하는 약)



DRUG DEVELOPMENT JOURNEY



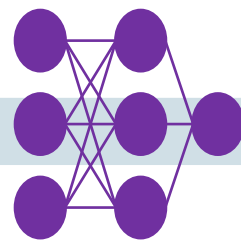
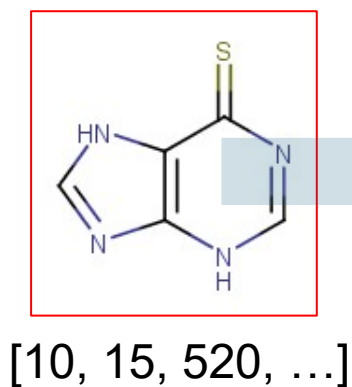
AI는 어떻게 활용될까?



약물 분자 구조를 숫자로...

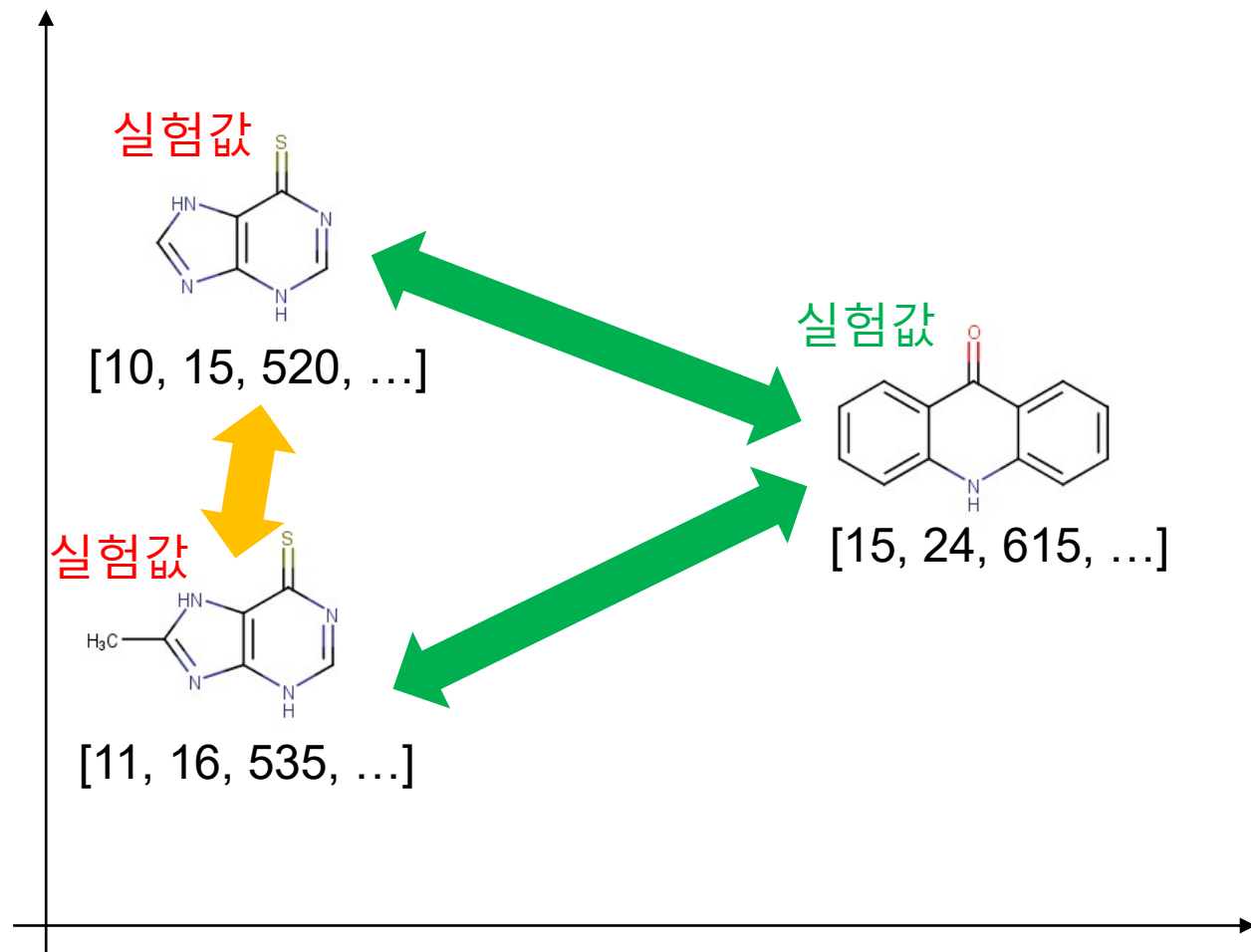
기본 가정:

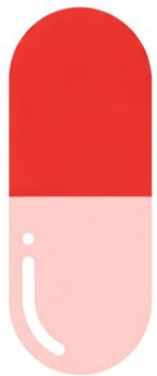
분자 구조가 유사하면 성질도 유사하다



Model

실험값

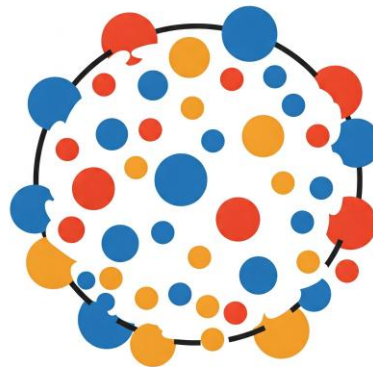




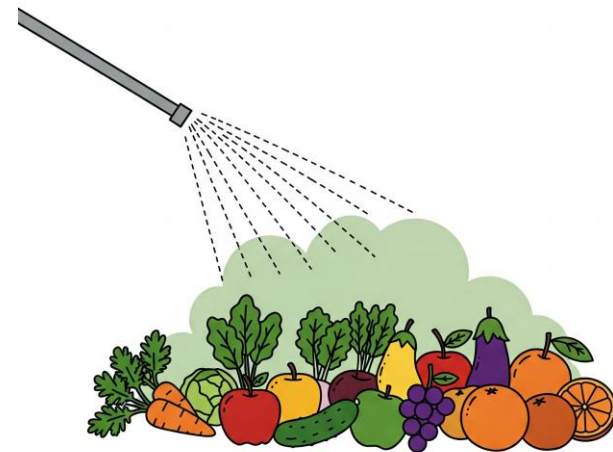
의약품



생활화학제품



나노 물질



농약



화장품

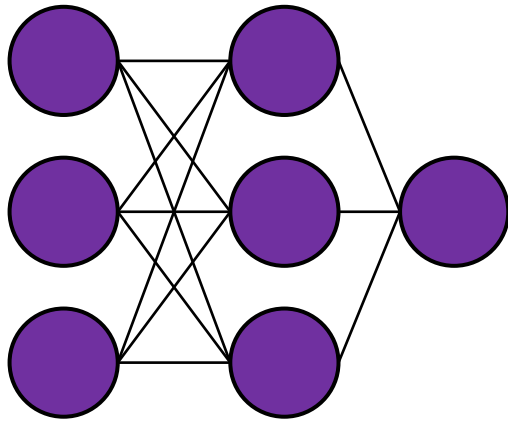


건강기능식품

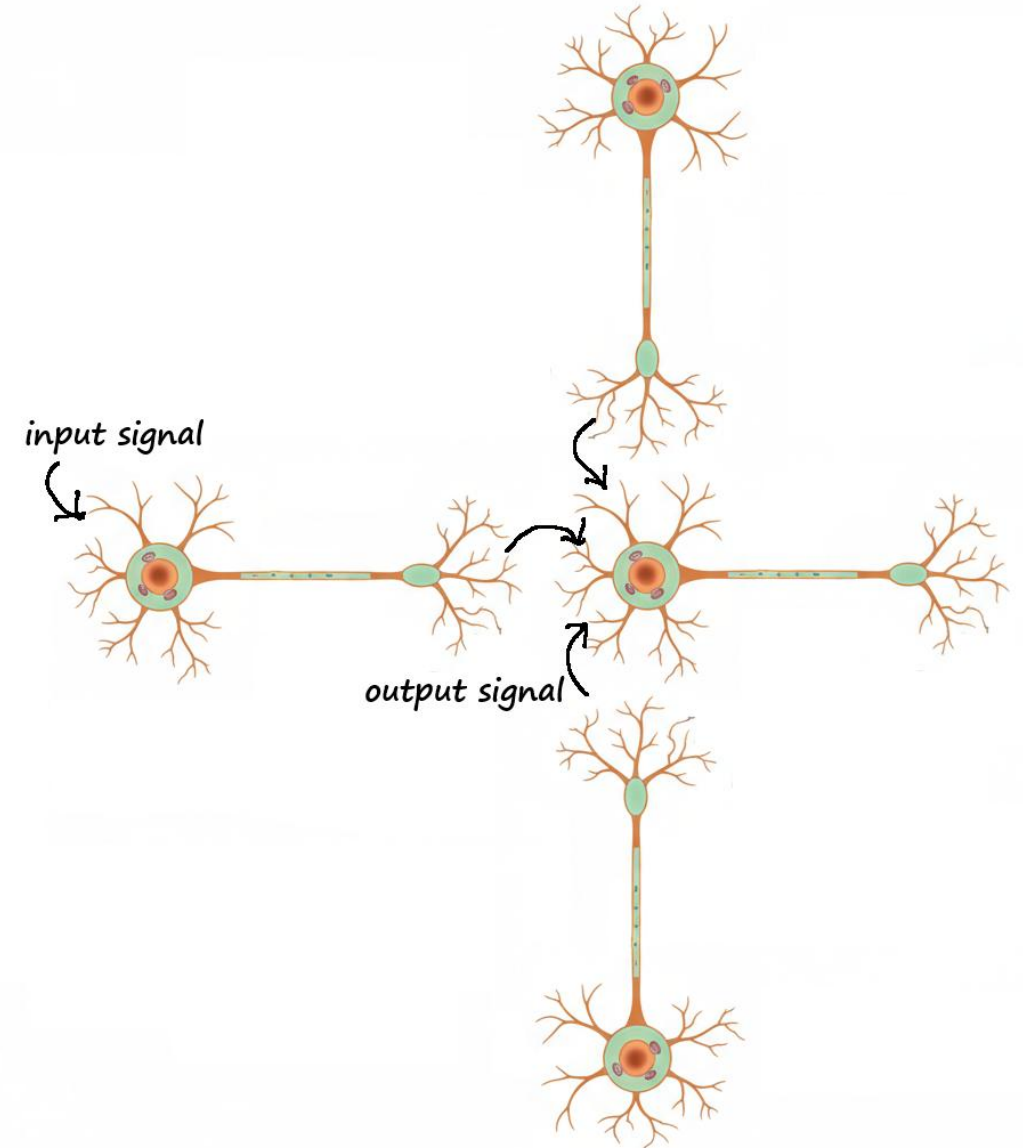


동물 의약품

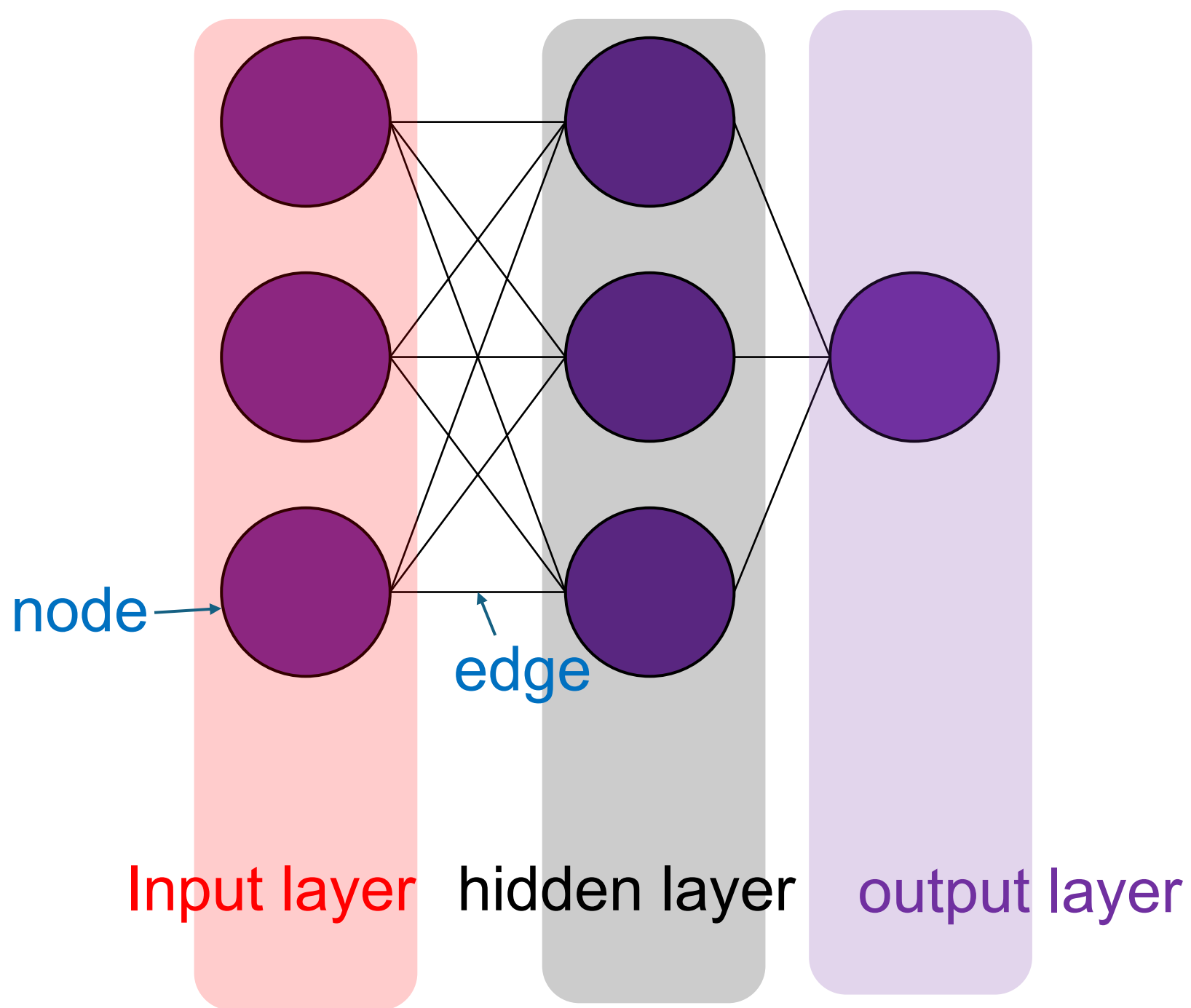
Biological neural network vs artificial neural network



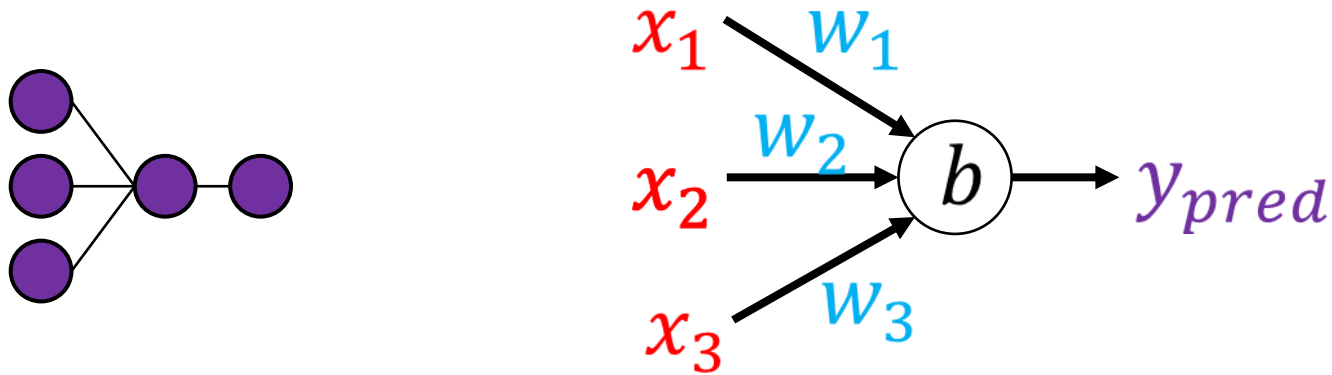
Analogy



- Architecture of AI model is defined after biological neural network
- Weight is a signal strength



Artificial neural network

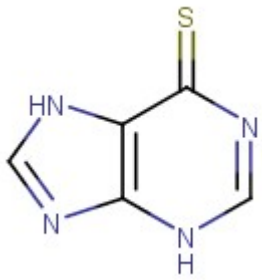


$$y_{pred} = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b$$

Feature vector: $[x_1, x_2, x_3]$

Weight/coefficient vector: $[w_1, w_2, w_3]$

Neural network for drug design

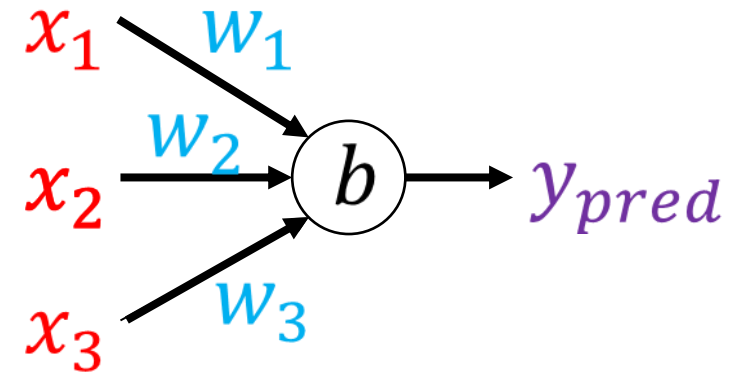


→ [10, 15, 520]

x_1 : 10

x_2 : 15

x_3 : 520

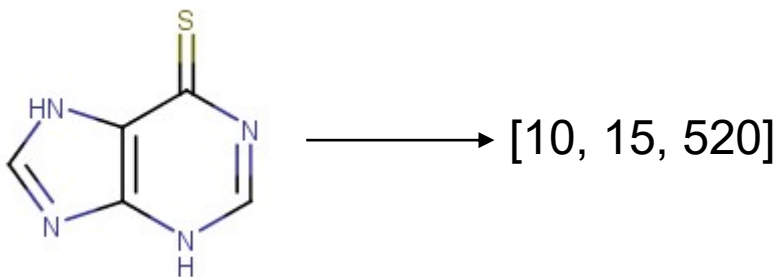


$$y_{pred} = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b$$

Feature vector: $[x_1, x_2, x_3]$

Weight/coefficient vector: $[w_1, w_2, w_3]$

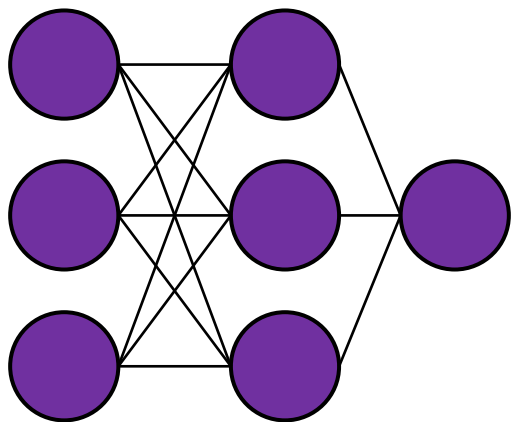
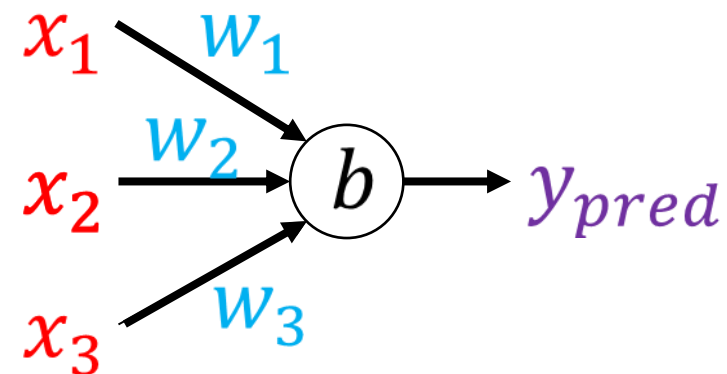
Neural network for drug design



$x_1: 10$

$x_2: 15$

$x_3: 520$

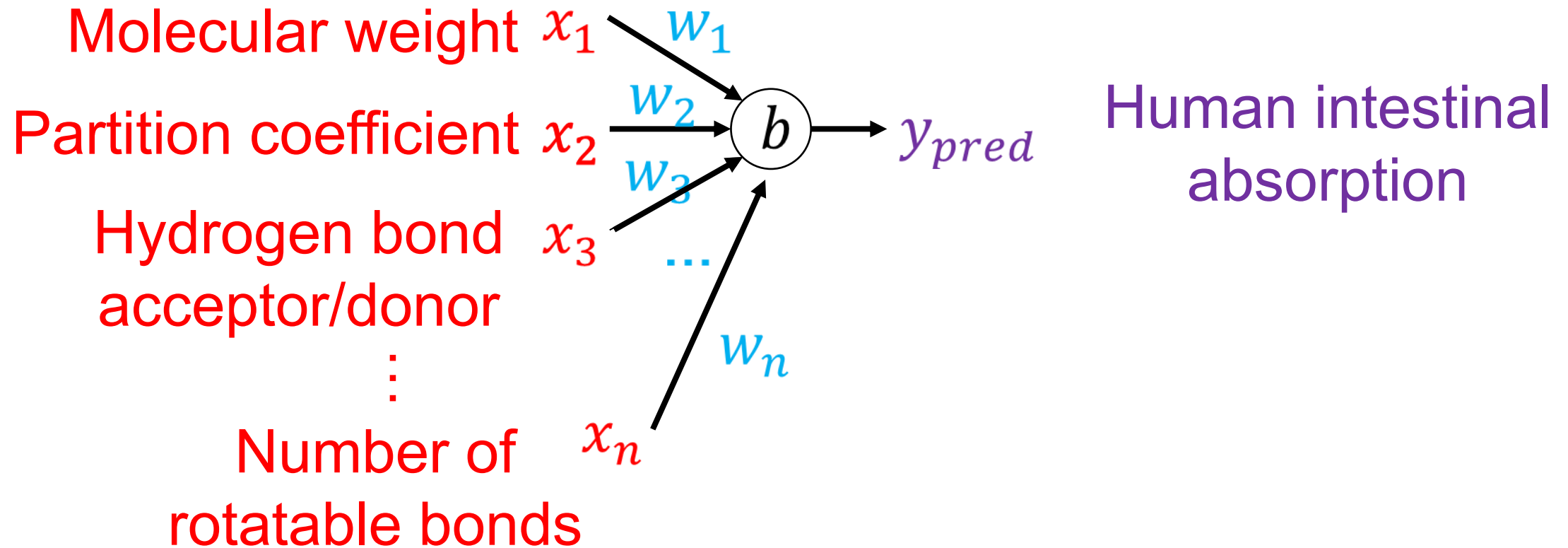


$$y_{pred} = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b$$

Feature vector: $[x_1, x_2, x_3]$

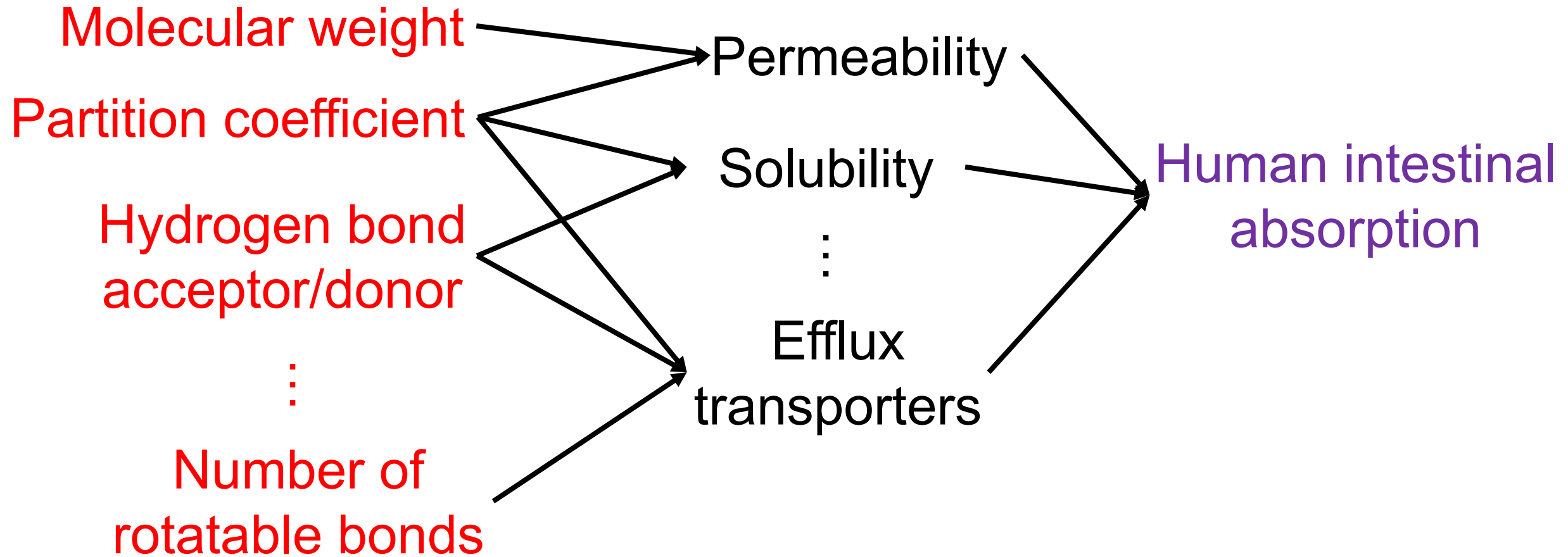
Weight/coefficient vector: $[w_1, w_2, w_3]$

Neural network for drug design

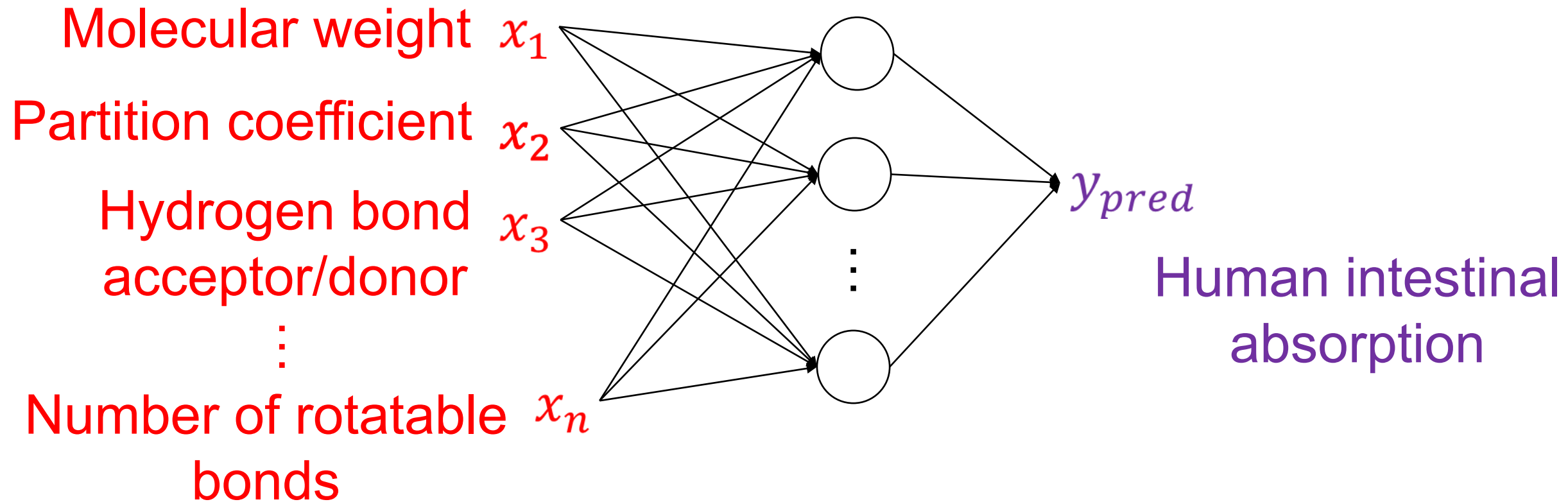


$$y_{pred} = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + \dots + w_n \cdot x_n + b$$

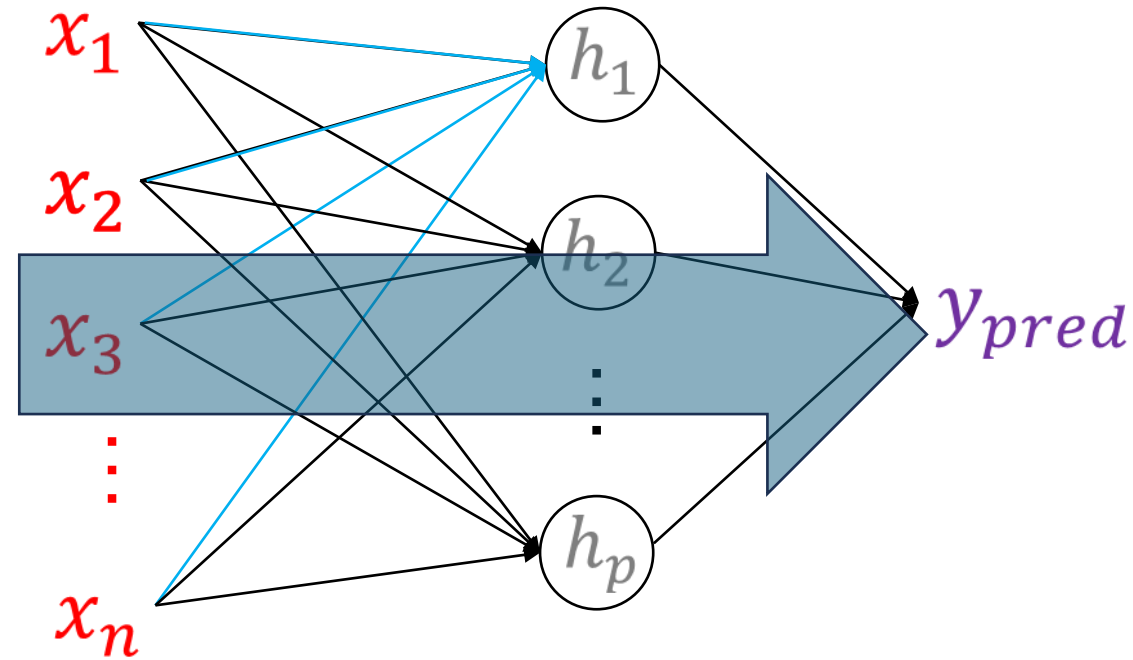
Neural network for drug design



Neural network for drug design



Neural network feed forward

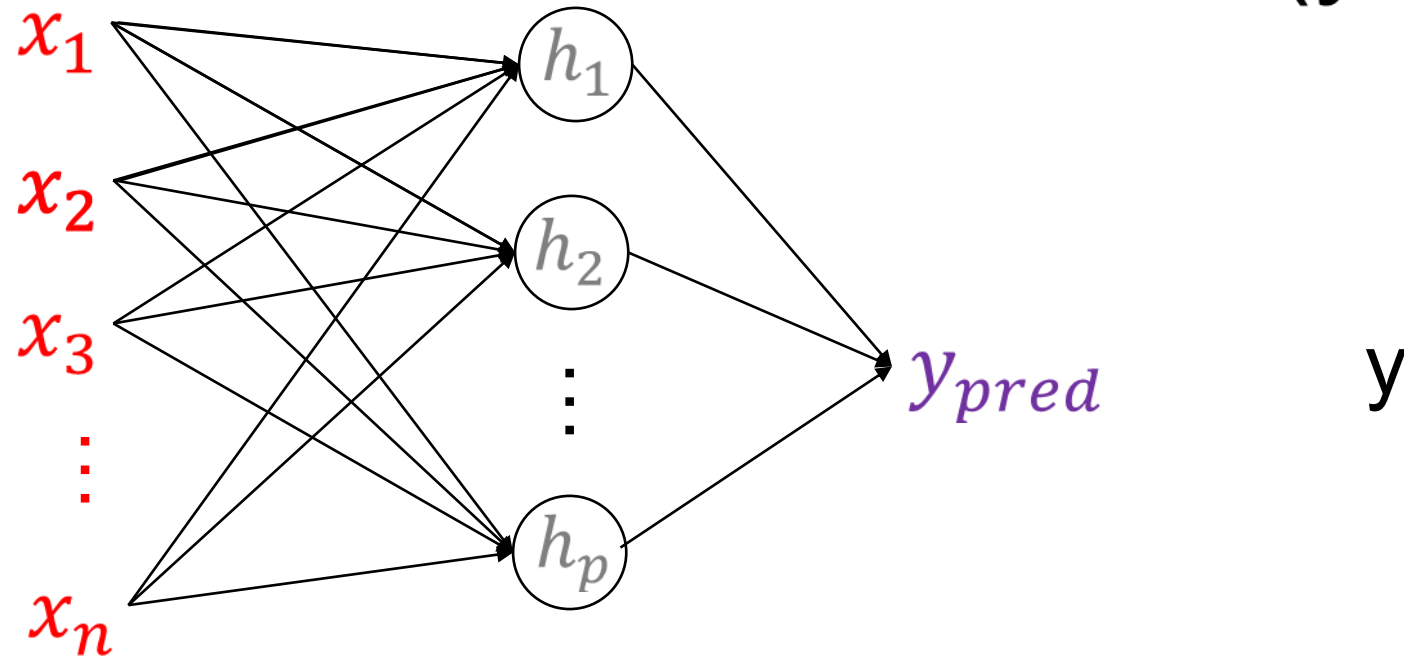


$$h_1 = w_{1,1} \cdot x_1 + w_{1,2} \cdot x_2 + \cdots + w_{1,n} \cdot x_n + b_1$$

Training process

Loss / Objective function

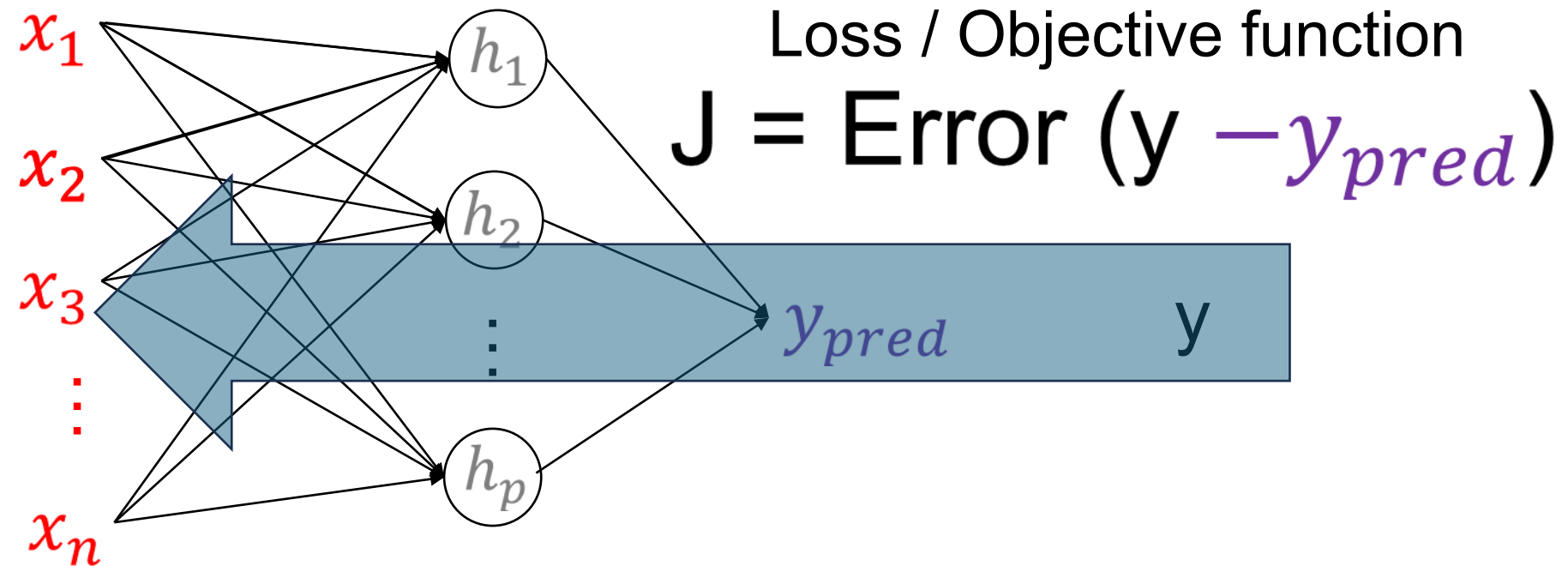
$$J = \text{Error} (y - y_{pred})$$



Weight: Random number (0~1)

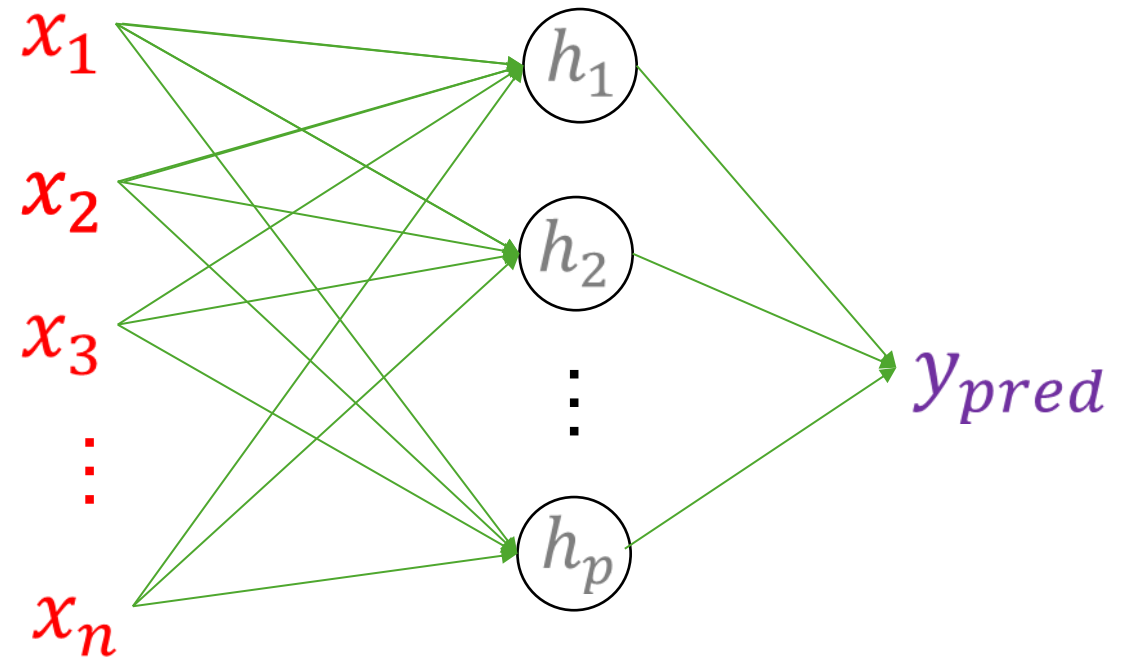
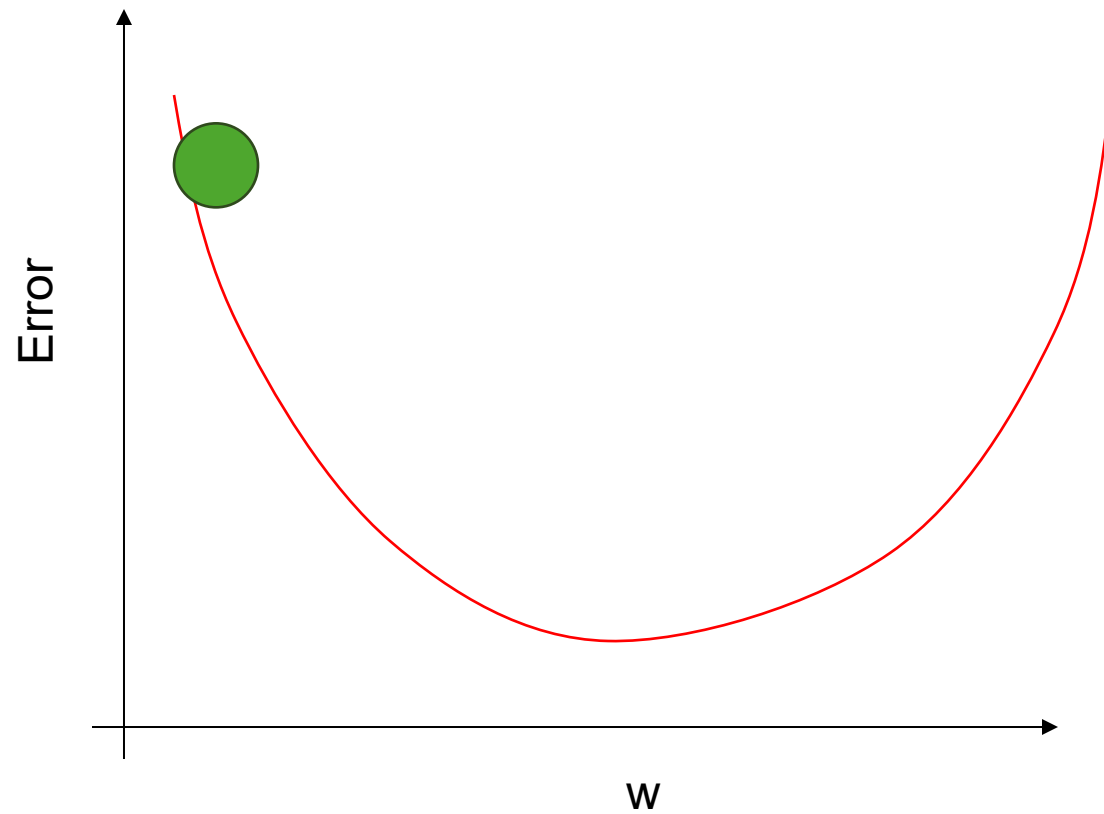
Training process: finding a set of weights where minimum error(J) was found.

Weight update (Backpropagation)

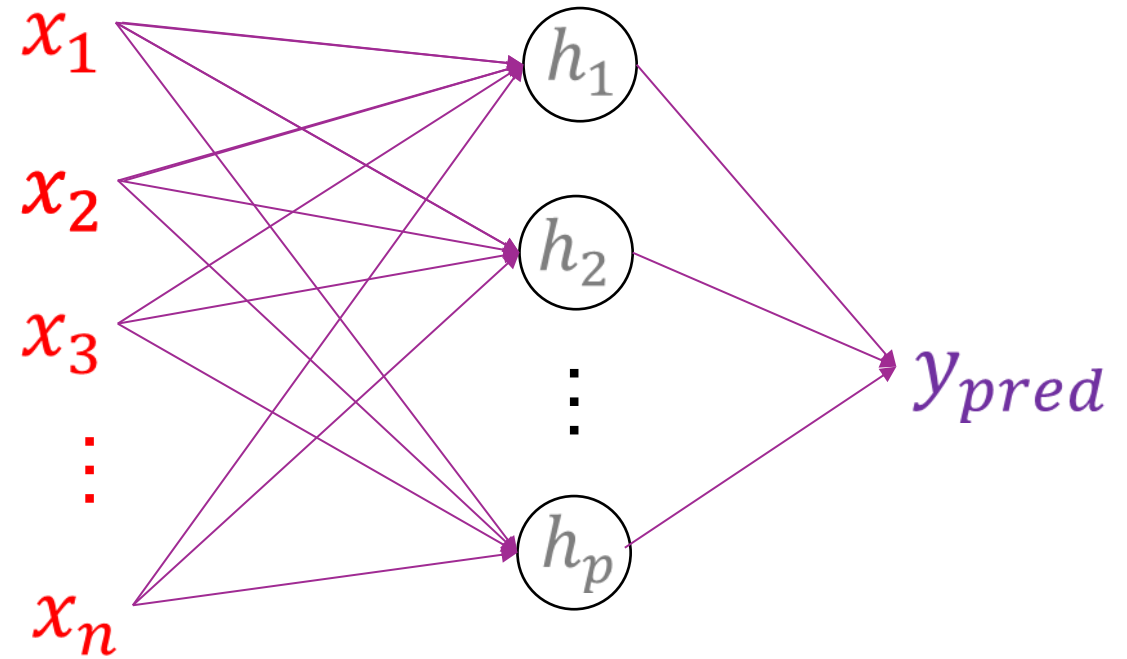
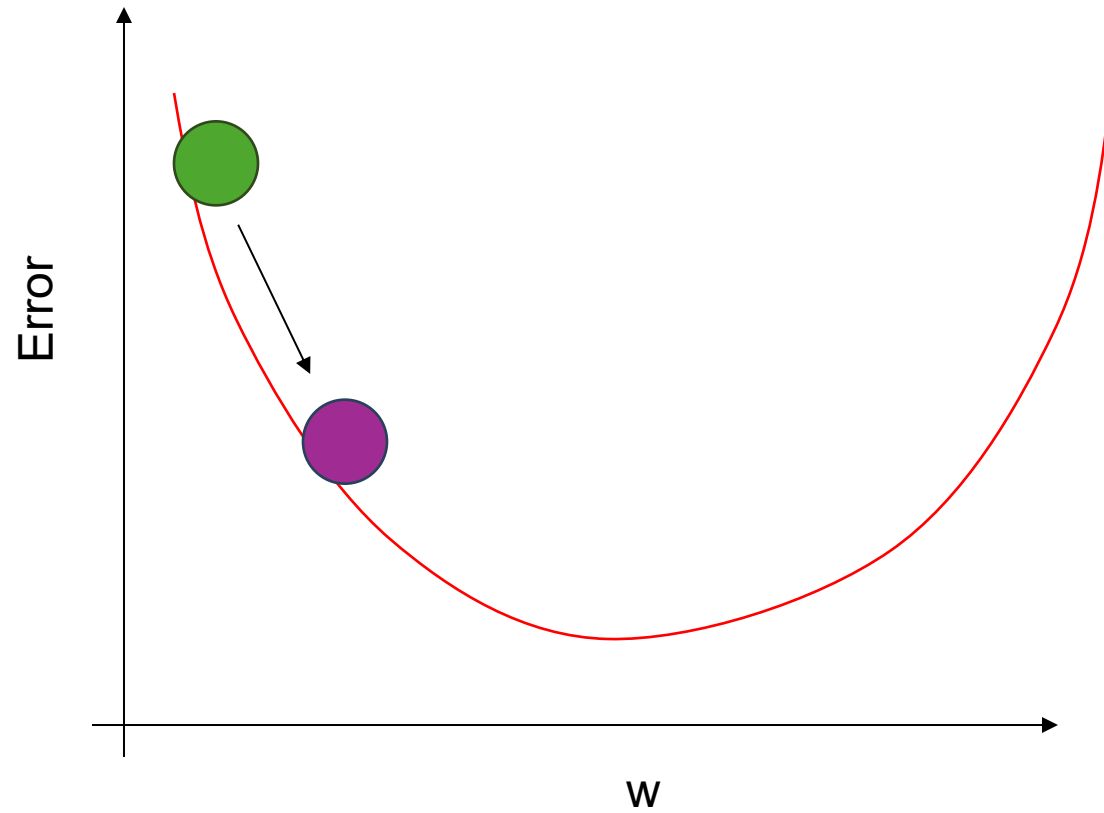


- 1) w is updated with differential value of the loss function
- 2) Repeat the updates until error is low enough

Weight update process



Weight update process



Weight update process

